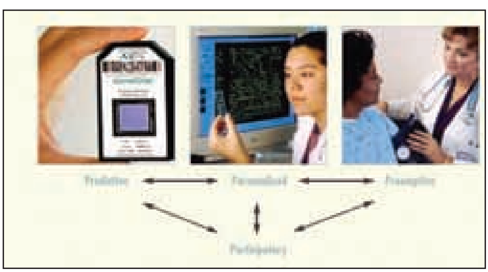
Newborn children will have their DNA quickly mapped, pointing out future potential problems, allowing us to curtail disease before it takes hold. Researchers have already successfully engineered a variety of scaffolds made from nanotubes and nanofibers that can be used to grow lifelike networks of cells from the liver, bladder, kidney, bones and cardiovascular system. These artificial tissues could be developed into new therapies for patients with diseased or damaged organs.

Further out into the future nanotechnology, can be expected to aid in the delivery of just the right amount of medicine to the exact spots of the body that need it most. Nanoshells, approximately 100nm in dia, will float through the body, attaching only to cancer cells. When excited by a laser beam, the nanoshells will give off heat — in effect, cooking the tumor and destroying it. Nanotechnology will create biocompatible joint replacements and artery stents that will last the life of the patient instead of having to be replaced every few years. Although the research is still exploratory, several groups of scientists are beginning to build novel nanostructures that mimic complex biomolecules. Some of these appear to have regenerative powers and might lead to therapies for untreatable conditions such as Alzheimer's, nerve injury and brain damage from stroke.


A suitable match?

Quite a few decades from now, it will probably become possible to create interfaces between the nervous system and electronic devices. These nanoscale links would make it possible to essentially "plug in" human-made machines to the body—for instance, prosthetic legs that truly function like one's own. With the knowledge of effectively stimulating muscles and advances in robotics and information science, it will become possible to generate intelligent walking properties. Interfaces between neurons and electronic devices would also make it possible to


The four revolutions